

EXERCISE 3 · MID DECK 2

Tool Use and Knowledge Bases

Apply the tool-use loop and RAG pipeline — including the classic gotchas.

6 questions · ~15–20 minutes · Worksheet only · answer key in separate PDF

WHAT TO DO

Six questions. About 15-20 minutes. Several involve reading code carefully — the bugs are subtle. If you've internalized the 8-step tool flow and the 5-step RAG pipeline, you'll catch them.

Q1 OF 6 · CONCEPT

In the 8-step tool-use flow, why does the deck emphasize that the model **never executes** the tool?

- A. Because the model would slow down if it had to run code.
- B. Because the model returns a structured request asking *your code* to run the tool, then your code sends the result back.
- C. Because AWS prohibits the model from making outbound network calls.
- D. Because tool execution requires a separate IAM permission the model doesn't have.

Your answer: _____

Q2 OF 6 · BUG SPOT

This tool-use code returns a valid response on the first API call but the second API call returns *ValidationException*. What did the developer forget?

```
if stop_reason == "tool_use":
    tool_use_block = response["output"]["message"]["content"][0]["toolUse"]
    result = run_tool(tool_use_block["name"], tool_use_block["input"])

    messages.append({
        "role": "user",
        "content": [{
            "toolResult": {
                "toolUseId": tool_use_block["toolUseId"],
                "content": [{"json": result}],
            }
        }]
    })

final = client.converse(
    modelId=model_id,
    messages=messages,
    toolConfig=TOOL_CONFIG,
)
```

Write your answer below:

Q3 OF 6 · CONCEPT

Why does semantic search find a match between *"refund window"* and *"customers can return items within 30 days"*, when keyword search does not?

- A. Semantic search uses regular expressions to find synonyms.
- B. Both phrases are converted to vectors by an embedding model; vectors with similar meaning are close in vector space.
- C. Bedrock maintains a built-in synonym dictionary that maps "refund" to "return".
- D. Semantic search uses fuzzy string matching with a Levenshtein distance threshold.

Your answer: _____

Q4 OF 6 · BUG SPOT

A learner is querying a Knowledge Base. The code below throws *AttributeError: 'BedrockRuntime' object has no attribute 'retrieve_and_generate'*. Why?

```
import boto3

client = boto3.client("bedrock-runtime", region_name="us-east-1")

response = client.retrieve_and_generate(
    input={"text": "When is spring break?"},
    retrieveAndGenerateConfiguration={
        "type": "KNOWLEDGE_BASE",
        "knowledgeBaseConfiguration": {
            "knowledgeBaseId": "ABC123",
            "modelArn": "us.amazon.nova-lite-v1:0",
        },
    },
)
```

Write your answer below:

Q5 OF 6 · CONCEPT

In the 5-step RAG pipeline, which steps happen **once during setup** and which happen **on every user query**?

- A. All five run on every query.
- B. Steps 1-3 (upload, embed, store) run once; steps 4-5 (retrieve, generate) run on every query.
- C. Steps 1 and 2 run once; steps 3-5 run on every query.
- D. All five run once during setup; nothing runs at query time.

Your answer: _____

Q6 OF 6 · DESIGN SCENARIO

Your team built a tool-use loop where the model can call `get_weather`, `get_stock_price`, and `send_email`.

Name **one risk** in this setup and **one mitigation** you'd add before shipping.

Write your answer below:
